

Notes CBPF IA school 2026: Power spectrum

Louis Legrand August 2026

1. Density fluctuations

The Universe is not perfectly homogeneous. We describe deviations from the mean density using the **density contrast**

$$\delta(\mathbf{x}, t) \equiv \frac{\rho(\mathbf{x}, t) - \bar{\rho}(t)}{\bar{\rho}(t)}.$$

- $\delta = 0$: mean density
- $\delta > 0$: overdensity
- $\delta < 0$: underdensity

The function $\delta(\mathbf{x}, t)$ contains the information about the distribution of matter.

As we saw yesterday, the evolution of the density contrast is, at linear order:

$$\ddot{\delta} + 2H\dot{\delta} - \left(\frac{c_s^2}{a^2} \nabla^2 + 4\pi G \bar{\rho}(t) \right) \delta = 0$$

It is thus very useful to pass into the Fourier space, where the spatial Laplacian ∇^2 transforms into $-k^2$. Thus each Fourier mode is evolving independantly.

We therefore decompose the density field into Fourier modes:

$$\delta(\mathbf{x}, t) = \int \frac{d^3k}{(2\pi)^3} \delta(\mathbf{k}, t) e^{i\mathbf{k} \cdot \mathbf{x}}.$$

with Fourier transform

$$\delta(\mathbf{k}, t) = \int d^3x \delta(\mathbf{x}, t) e^{-i\mathbf{k} \cdot \mathbf{x}}.$$

Note that in cosmology, the Fourier transform convention often uses the $e^{-i\mathbf{k} \cdot \mathbf{x}}$ instead of the $e^{-2i\pi\mathbf{k} \cdot \mathbf{x}}$, thus there is an additional normalization in the inverse FT to take into account.

With this convention, $k = \frac{2\pi}{\lambda}$, where λ is the wavelength of the mode. Therefore: - small $k \rightarrow$ large spatial scales - large $k \rightarrow$ small spatial scales

The evolution equation becomes

$$\ddot{\delta}_{\mathbf{k}} + 2H\dot{\delta}_{\mathbf{k}} + \left(\frac{c_s^2 k^2}{a^2} - 4\pi G \bar{\rho} \right) \delta_{\mathbf{k}} = 0.$$

Notice that the equation for a given $\mathbf{k}$ contains **only that same $\mathbf{k}$** . There is no $\delta_{\mathbf{k}'}$ with $\mathbf{k}' \neq \mathbf{k}$. Hence each *Fourier mode evolves independently*.

2. From Fourier modes to the power spectrum

We cannot predict the exact value of each density mode in the Universe today, but we can make a statistical prediction.

Thus, for a statistically isotropic sky, we define the **matter power spectrum** through

$$\langle \delta(\mathbf{k}) \delta^*(\mathbf{k}') \rangle = (2\pi)^3 \delta_D(\mathbf{k} - \mathbf{k}') P(k).$$

The matter power spectrum tells us how much variance in the matter density field exists at each spatial scale.

Shape of the power spectrum

Since all scales evolve independently, we can write

$$\delta(k, z) = D(z) \delta_{\text{initial}}(k)$$

where D is called the growth function.

Then we have

$$P(k, z) = D^2(z) T^2(k) P(k)_{\text{initial}}$$

Where $P(k)_{\text{initial}}$ is the initial power spectrum at the end of inflation, and $T(k)$ is the transfer function, and encodes how each mode evolve. This evolution depends when did the mode enter the orizon, if they enter during the matter or during the radiation domination.

The turnover scale in the $P(k)$ corresponds to the scale of the horizon at the transition between radiation and matter domination. At higehr k the modes where inside the horizon, so they started to oscillate, which damped the power spectrum. At smaller k , the modes are still outside the horizon, they didnt evolve and are directly following the shape of the intial matter power spectrum at the end of the inflation.

3. Spherical harmonics

Even if the Universe is 3D, we naturally observe galaxies in a sphere, over the 2D sky. We thus define the galaxy overdensity field, defined by the unit vector $\hat{n}$ and the redshift z : $\delta_g(\hat{n}, z)$

The equivalent of the Fourier modes on a sphere are the spherical harmonics $Y_{\ell m}$, where the multipole number $\ell \in \mathbb{N}$ and the azimuthal number $m \in [-\ell, \dots, \ell]$. Therefore there are $2\ell + 1$ different m -modes for each ℓ .

You might have seen the spherical harmonics in quantum mechanics.

Similarly to the FT, we decompose the density fields into its spherical harmonic coefficients $a_{\ell m}$

$$\delta_g(\hat{n}) = \sum_{\ell, m} a_{\ell m} Y_{\ell m}(\hat{n}),$$

where

$$a_{\ell m} = \int d\Omega \delta_g(\hat{n}) Y_{\ell m}^*(\hat{n}).$$

In practice, $\ell = 0$ is called the monopole, and measures the mean overdensity, $\ell = 1$ is called the dipole, $\ell = 2$ is the quadrupole, etc. . .

Similarly to the fourier wave number k , the multipole ℓ is the inverse of the angular scale θ , and roughly

$$\ell \sim \frac{180^\circ}{\theta}$$

Therefore:

- small $\ell \rightarrow$ large angular scales
- large $\ell \rightarrow$ small angular scales

For example $\ell = 100$ is more or less a scale of 1,8 degrees on the sky.

4. Angular power spectrum

We now do essentially the same thing we did for $P(k)$. For an isotropic and homogeneous field, we have

$$\langle a_{\ell m} a_{\ell' m'}^* \rangle = C_\ell \delta_{\ell \ell'} \delta_{m m'}.$$

where we introduced the angular power spectrum C_ℓ .

The power spectrum therefore depends only on ℓ , not on m .

4.1 Estimator

From observations of the galaxy overdensity field, we can define an estimator for the angular power spectrum

$$\hat{C}_\ell = \frac{1}{2\ell + 1} \sum_{m=-\ell}^{\ell} |a_{\ell m}|^2.$$

This estimator is unbiased in the sense that $\langle \hat{C}_\ell \rangle = C_\ell$. However, the estimator has a nonzero variance corresponding to an irreducible error in our determination of the true power spectrum

$$\frac{\Delta C_\ell}{C_\ell} = \frac{\sqrt{\langle (C_\ell - \hat{C}_\ell)^2 \rangle}}{C_\ell} = \sqrt{\frac{2}{2\ell + 1}}$$

As expected, this is worse for low multipoles, as there is only a few m 's that we can average upon to get an estimate for C_ℓ .

4.2 Comoving distance

The comoving distance is the distance between two objects in an expanding universe after factoring out the expansion of the universe. For an object observed at redshift z , the radial comoving distance is

$$\chi(z) = c \int_0^z \frac{dz'}{H(z')}.$$

4.3 Galaxy power spectrum

Suppose we bin the galaxies in a redshift slice, defined by $W(\chi)$, where χ is the comoving distance. Then, the overdensity of galaxies in the redshift slice is given by

$$\delta_g(\hat{n}) = \int d\chi W(\chi) \delta_g(\chi \hat{n}, \chi).$$

From there we can obtain the expression of the galaxy angular power spectrum as a function of the 3D matter power spectrum $P(k)$

$$C_\ell = \frac{2}{\pi} \int dk k^2 P(k) \left[\int d\chi W(\chi) j_\ell(k\chi) \right]^2,$$

where $j_\ell(k\chi)$ is a spherical Bessel function. It describes how a 3D Fourier mode projects onto a spherical harmonic with multipole ℓ .

4.4 Limber approximation

For large $k\chi$, the spherical Bessel functions can be approximated by a Dirac δ . We thus obtain a simplified expression for the galaxy power spectrum

$$C_\ell \simeq \int \frac{d\chi}{\chi^2} W^2(\chi) P\left(k = \frac{\ell + 1/2}{\chi}\right).$$

5. Galaxies are biased tracers of matter

Galaxies do not perfectly trace the matter density, we say that they are a biased tracer. On sufficiently large scales, we can approximate it with a linear bias defined by

$$\delta_g = b_g \delta_m$$

where b_g is the galaxy bias.

Therefore,

$$P_g(k, z) = b_g^2(z) P_m(k, z)$$

The Limber expression for the galaxy angular power spectrum becomes

$$C_\ell^{gg} \simeq \int d\chi \frac{W^2(\chi)}{\chi^2} b_g^2(z) P_m\left(\frac{\ell + 1/2}{\chi}, z\right)$$

This is the standard equation connecting the theoretical matter power spectrum to an observed galaxy angular power spectrum.